

Mathematics

By a group of supervisors

PARENTS' GUIDE

EL-MOASSER Interactive E-learning Application



FIRST TERM

4th
PRIMARY

General notes for parents	2
How to use this guide ?	8
Revision	10

UNIT 1 Place Value

Concept 1 Reinforcing Place Value

Lessons 1&2	Big Numbers! – Changing Values	16
Lessons 3&4	Many Forms to Write Numbers – Composing and Decomposing	26

Concept 2 Using Place Value

Lessons 5&6	Comparing Big Numbers – Comparing Numbers in Multiple Forms	35
Lesson 7	Descending and Ascending Numbers	41
Lesson 8	Rounding Rules	47



UNIT 2 Addition and Subtraction Strategies

Concept 1 Using Addition and Subtraction Strategies

Lesson 1	Properties of Addition	56
Lesson 2	Addition with Regrouping	62
Lesson 3	Subtraction with Regrouping	68

Concept 2 Solving Multistep Problems

Lesson 4	Bar Models, Variables and Story Problems	77
Lesson 5	Solving Multistep Story Problems with Addition and Subtraction	85



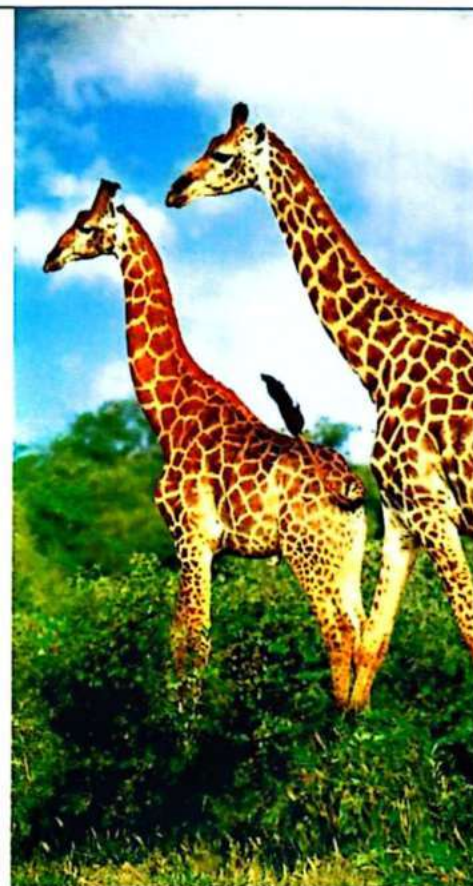
UNIT 3 Concepts of Measurement

Concept 1 Metric Measurement

Lesson 1	Measuring Length	92
Lesson 2	Measuring Mass	97
Lesson 3	Units of Capacity	103

Concept 2 Measuring Time

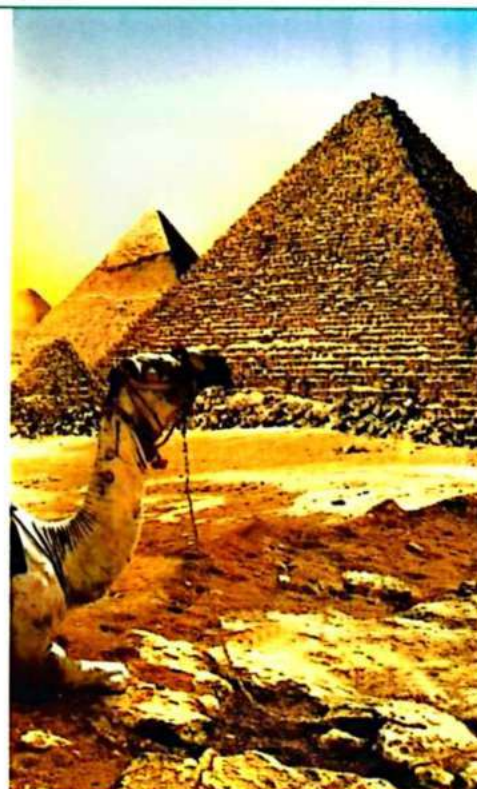
Lessons 4&5	Units of Time - Elapsed Time	110
Lesson 6	Applications of Measurement 1	122
Lesson 7	Applications of Measurement 2	129



UNIT 4 Area and Perimeter

Concept 1 Explore Area and Perimeter

Lesson 1	Finding Perimeter	138
Lesson 2	Finding Area	145
Lesson 3	Unknown Dimensions	152
Lesson 4	Complex Shapes	159



UNIT 5 Multiplication as a Relationship

Concept 1 Multiplicative Comparisons

Lesson 1 Multiplicative Comparison 168

Lessons 2&3 Creating Multiplicative Comparison Equations -
Solving Multiplicative
Comparison Equations 173

Concept 2 Properties and Patterns of Multiplication

Lessons 4&5 Commutative Property of Multiplication -
Identity Property and the Zero Property 182

Lessons 6&7 Associative Property of Multiplication -
Applying Patterns in Multiplication 188



UNIT 6 Factors and Multiples

Concept 1 Understanding Factors

Lessons 1&2 Identifying Factors of Whole Numbers -
Prime and Composite Numbers 200

Lesson 3 Greatest Common Factor (G.C.F) 210

Concept 2 Understanding Multiples

Lessons 4&5 Identifying Multiples of Whole Numbers -
Common Multiples 218

Lesson 6 Relationships between Factors
and Multiples 226



UNIT 7

Multiplication and Division : Computation and Relationships

Concept 1	Multiplying by 1-Digit and 2-Digit Factors	
Lessons 1&2	The Area Model Strategy - The Distributive Property	234
Lessons 3&4	The Partial Products Algorithm - Multiply by a One-Digit Number	242
Lesson 5	Multiply a Two-digit Number by a Multiple of 10	252
Concept 2	Dividing by 1-Digit Divisors	
Lesson 6	Exploring Remainders	258
Lesson 7	Patterns in Division	263
Lesson 8	The Area Model and Division	268
Lessons 9&10	The Partial Quotients Algorithm - The Standard Division Algorithm	273
Lesson 11	Division and Multiplication	283



UNIT 8

Order of Operations

Concept 1	Order of Operations	
Lessons 1&2	Order of Operations - The Order of Operations and Story Problems	292

Glossary

301



REVISION

In this revision your child will review on what he/she had learned in primary three.

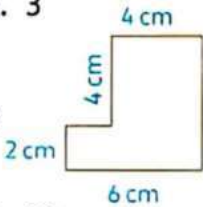


Revision 1

1. Complete.

- $32,621 + 18,709 =$
- 30 thousands = hundreds.
- The perimeter of the rectangle  is cm.
- The place value of the digit 4 in the number 46,385 is
- Thirty-eight thousand, five hundred two in standard form is

2. Choose the correct answer.

- $35 \div 7 =$
A. 5 B. 6 C. 7 D. 8
- $3,000 + 829 =$
A. 3,829 B. 8,293 C. 30,829 D. 3,928
- $8 \times \quad = [8 \times 5] + [8 \times 2]$
A. 10 B. 3 C. 8 D. 7
- The area of the shape  is square cm.
A. 30 B. 28 C. 24 D. 20
- The greatest number formed from 7, 2, 0, 6, 8, 1 is
A. 870,621 B. 876,210 C. 102,678 D. 780,621

3. Arrange the following numbers from least to greatest.

56,210 , 506,021 , 650,201 , 171,000 , 43,692

The order is : , , , ,

4. Find.

a.
$$\begin{array}{r} 7,263 \\ - 4,081 \\ \hline \end{array}$$

b.
$$\begin{array}{r} 43,826 \\ + 25,095 \\ \hline \end{array}$$

c. $9 \times 3,000 =$

d. $7 \times 7 =$

Revision 2

1. Choose the correct answer.

a. $54,275 - 32,938 =$ _____

A. 12,337

B. 21,373

C. 21,733

D. 21,337

b. $501,326 <$ _____

A. 510,200

B. 501,236

C. 51,623

D. 56,632

c. $3 \times 80 =$ _____

A. 24

B. 240

C. 2,400

D. 24,000

d. $\frac{1}{7}$ of 28 = _____

A. $\frac{1}{8}$ of 32

B. $\frac{1}{5}$ of 30

C. $\frac{1}{6}$ of 48

D. $\frac{1}{9}$ of 18

e. The perimeter of the square  is _____ cm.

A. 20

B. 25

C. 10

D. 30

2. Complete.

a. Eight hundred sixty-three thousands, five hundred seven in standard form is _____

b. The place value of the digit 7 in 762,435 is _____

c. $7 \times$ _____ $= 63$

d. _____ $= 100,000 + 7,000 + 30 + 5$

e. _____ tens $= 170$

3. A factory produces 800 cans of soft drink every day.

How many cans the factory produces in a week ?

4. Write the greatest number and smallest number can be formed

from 9, 4, 0, 3, 1, 6.

- The greatest number : _____

- The smallest number : _____



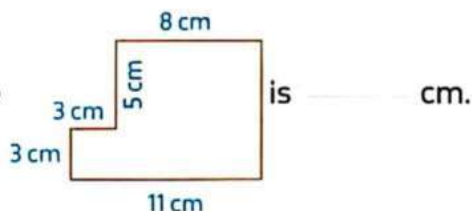
Revision 3

1. Complete.

a. $\quad \div 4 = 8$

b. $[5 \times 6] \times 7 =$

c. The perimeter of the shape



is $\quad$ cm.

d. $8,762 - 7,648 =$

e. The smallest number formed from 2, 7, 0, 6, 5 is

2. Choose the correct answer.

a. The value of the digit 3 in 721,362 is

A. 30,000

B. 3,000

C. 300

D. 30

b. The area of the rectangle whose length is 10 cm and width is 7 cm is $\quad$ square cm.

A. 17

B. 34

C. 70

D. 44

c. $\quad = 1,000 + 900 + 70 + 2$

A. 19,472

B. 1,927

C. 10,972

D. 1,972

d. $61,072 + 9,838 =$

A. 7,910

B. 70,910

C. 79,010

D. 70,091

e. $6 + 6 + 6 + 6 + 6 = 6 \times$

A. 6

B. 5

C. 7

D. 8

3. Arrange the following from least to greatest.

5×12 , 7×8 , 3×10 , 6×9 , 8×1

The order is: $\quad$, $\quad$, $\quad$, $\quad$, $\quad$

4. Bassem has 72 marbles, he wants to put each 8 marbles in a bag.

How many bags does Bassem need ?

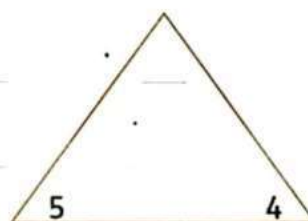
5. Find the product. Write the fact family.

$\quad \times \quad = \quad$

$\quad \times \quad = \quad$

$\quad \div \quad = \quad$

$\quad \div \quad = \quad$



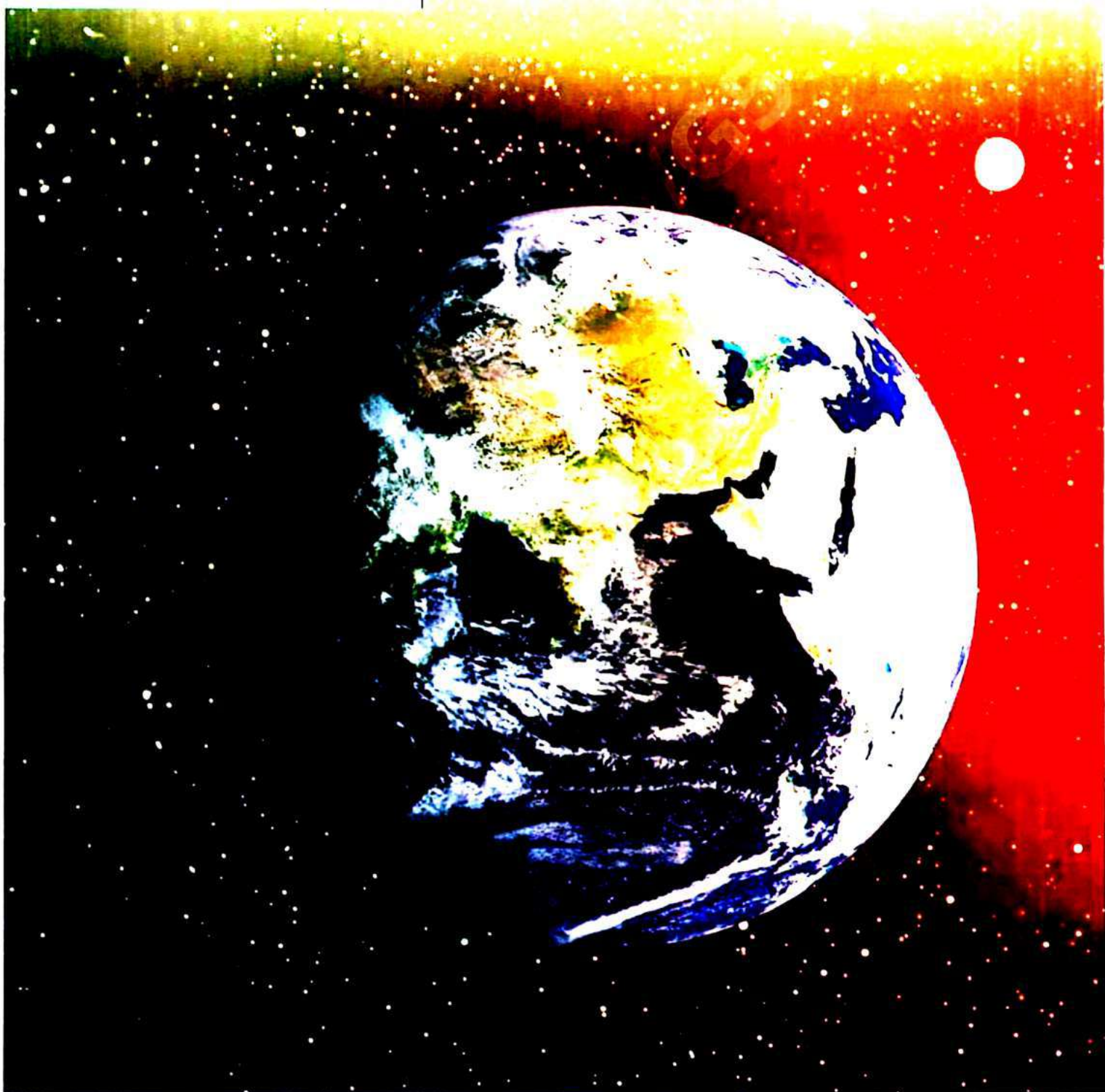
THEME ONE

Number Sense and Operations

UNIT 1

Place Value

- ▶ **Concept 1 :**
Reinforcing Place Value
- ▶ **Concept 2 :**
Using Place Value



CONCEPT

1

Reinforcing Place Value

► Lessons 1&2

Big Numbers!
Changing Values

Learning Objectives:

- Students will identify all whole number place values through the One Milliard place.
- Students will explain how the value of a digit changes based on its place in a number
- Students will explain how the value of a digit changes as it moves to the left in a whole number.
- Students will describe patterns they observe in changing place values.

► Lessons 3&4

Many Forms to Write Numbers
Composing and Decomposing

Learning Objectives:

- Students will write numbers in standard, word, and expanded forms .
- Students will compose and decompose number in multiple forms .

Fast Fact

- The distance between the Earth and the Sun is about **149,598,000 km.**
- The least distance from the Earth to the Moon is about **384,402 km.** which equals **384,402,000 m.**

Lessons 1 & 2

► Big Numbers! ► Changing Values

Remember The place value

The value of each digit in any number depends on its place in this number.

For Example :

The least distance from the Earth to the moon is 384,402 km.

Notice the value of each digit in the number 384,402.

Note that :

Each group of three digits is called a period. Each period has ones, tens and hundreds in it.

PERIOD			PERIOD		
THOUSANDS			ONES		
Hundreds	Tens	Ones	Hundreds	Tens	Ones
3	8	4	4	0	2

3 8 4 , 4 0 2

3 is in the hundred thousands place. Its value is 300,000

8 is in the ten thousands place. Its value is 80,000

4 is in the thousands place. Its value is 4,000

4 is in the hundreds place. Its value is 400

0 is in the tens place. Its value is 0

2 is in the ones place. Its value is 2

Example 1

Write the place value and the value of the colored digit.

- | | place value | value |
|---------------------|----------------------|----------------------|
| a. 7 9 , 285 | <input type="text"/> | <input type="text"/> |
| b. 980,7 5 8 | <input type="text"/> | <input type="text"/> |
| c. 2 9,510 | <input type="text"/> | <input type="text"/> |
| d. 220,0 8 9 | <input type="text"/> | <input type="text"/> |

Solution

- | | place value | value |
|----|---------------|--------|
| a. | thousands | 9,000 |
| b. | tens | 50 |
| c. | ten thousands | 20,000 |
| d. | hundreds | 0 |

Notes for parents :

- Let your child remember that the position of a digit in a number determines its value.

Learn 1 Really big numbers

Million

- You know that the greatest 6-digit number is 999,999.

The number which comes just after 999,999 is **1,000,000**. It is read as **one million** which is the smallest 7-digit number.

- To show **1,000,000** in the place-value chart, a period for **Millions** has to be added to the left of the Thousands period.

PERIOD			PERIOD			PERIOD		
MILLIONS			THOUSANDS			ONES		
Hundreds	Tens	Ones	Hundreds	Tens	Ones	Hundreds	Tens	Ones
		1	0	0	0	0	0	0

Written as : 1,000,000

Read as : One million

More about millions

Egypt population in 2020 was 102,334,404
look at this number on the place-value chart.



PERIOD			PERIOD			PERIOD		
MILLIONS			THOUSANDS			ONES		
H	T	O	H	T	O	H	T	O
1	0	2	3	3	4	4	0	4
↑	↑	↑	↑	↑	↑	↑	↑	↑
Hundred Millions place	Ten Millions place	Millions place	Hundred Thousands place	Ten Thousands place	Thousands place	Hundreds place	Tens place	Ones place

This number read as :

One hundred two million, three hundred thirty-four thousand, four hundred four.

or in a short way : 102 million, 334 thousand, 404

Math tip

The place-value chart helps you read greater numbers. You say : "102" then at the comma you name the period, "million".

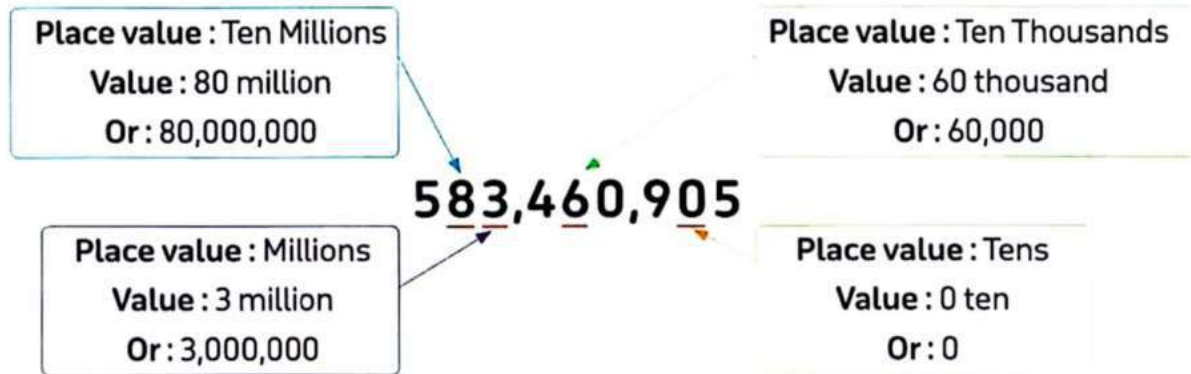
- Help your child apply and extend understanding of the place value system to multi-digit whole numbers.

Example 2

What is the place value and the value of each underlined digit ?

583,460,905

Solution



Milliard (Billion)

China has the world's largest population. In 1980, the population of China reached about **1,000,000,000**. It is read as **one milliard** [or **one billion**].

To show 1,000,000,000 in the place-value chart, a column for **Milliards** has to be added to the left of the Millions period.



PERIOD			PERIOD			PERIOD			
MILLIARDS	MILLIONS			THOUSANDS			ONES		
Ones	Hundreds	Tens	Ones	Hundreds	Tens	Ones	Hundreds	Tens	Ones
1	0	0	0	0	0	0	0	0	0

Written as : 1,000,000,000

Read as : One milliard

Notes for parents :

- Ask your child to tell you the value and the place value of each digit in the number : 243,019,507.

More about millions

The world's population in 2020 was about 7,794,798,739

Look at this number on the place-value chart.

PERIOD			PERIOD			PERIOD			
MILLIARDS	MILLIONS			THOUSANDS			ONES		
0	H	T	O	H	T	O	H	T	O
7	7	9	4	7	9	8	7	3	9

This number read as :

Seven milliard, seven hundred ninety-four million,
seven hundred ninety-eight thousand,
seven hundred thirty-nine.

or in a short way :

7 milliard, 794 million, 798 thousand, 739



Example 3

In the number 3,418,079,265, what digit is in the :

- Thousands place ?
- Ten Millions place ?
- Milliards place ?
- Hundred Thousands place ?

Solution

- 9
- 1
- 3
- 0

How to read a large number ?

- Divide the number [from right to left] into "periods" each period contains 3 digits.

6,208,196,318

- Use the place-value chart to help you read the large number.

PERIOD			PERIOD			PERIOD			
MILLIARDS	MILLIONS			THOUSANDS			ONES		
0	H	T	O	H	T	O	H	T	O
6	2	0	8	1	9	6	3	1	8
6 milliard	208 million			196 thousand			318		

• Help your child use periods to read multi-digit numbers in an easy way.

- 3 Start from the left and read the number in each period followed by the period name as follows.

Reading

6,208,196,318

Six milliard, two hundred eight million,
one hundred ninety-six thousand, three hundred eighteen.

In a short way : 6 milliard, 208 million, 196 thousand, 318

Example 4

Choose the correct answer.

- 4 milliard, 103 million, 905 thousand, 484 =
A. 43,509,458 B. 403,590,548 C. 4,103,905,484 D. 4,950,854
- Forty-three million, five hundred nine thousand, four hundred fifty-eight =
A. 43,509,458 B. 403,590,548 C. 4,103,905,484 D. 4,950,854
- 403 million, 590 thousand, 548 =
A. 43,509,458 B. 403,590,548 C. 4,103,905,484 D. 4,950,854
- The value of the digit 3 in the Hundreds place is
A. 3 B. 30 C. 300 D. 3,000

Solution

1. C 2. A 3. B 4. C

Check your understanding

- In each of the following numbers.
 - underline the digit in the Hundred Thousands place.
 - circle the digit in the Ten Millions place.
 - draw a square around the digit in the Millions place.

a. 7,561,492,048 b. 3,914,500,721
- Read the following numbers.

a. 912,031,301 b. 70,804,230 c. 5,003,521,216
- What is the value of each of the following.

a. 8 in the Tens place ? b. 5 in the Ten Thousands place ?

Notes for parents :

- Ask your child to write a number through milliard and then ask him/her to read it loudly.

Learn 2 Changing place values

- The value of a digit changes as it moves to the left within a number.
- Our place-value system is based on **tens**.

Each place value in this system is **10 times** the one to the right of it.



	$\times 10$	$\times 10$	$\times 10$	$\times 10$	$\times 10$	$\times 10$
Millions	Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
2	2	2	2	2	2	2
2,000,000	200,000	20,000	2,000	200	20	2

- A digit in one place equals 10 times the digit in the place to its right.
For Example: The Hundreds place is 10 times the Tens place, so the value of 2 changes from 20 to 200.
- Observe the pattern in the number of zeroes.

Example 5

Fill in the blanks below.

- The value of the digit 3 in the number 7,431,210 equals _____ times the value of the digit 3 in the number 4,563,809
- The value of 7 in the Thousands place = _____ times the value of 7 in the Tens place.
- _____ is 10 times one hundred thousand.
- 30 tens equals _____
- 7,000 thousands = _____ millions.

Solution

- 10
- 100
- 1,000,000
- 300 [**think** : $30 \times 10 = 300$]
- 7 [**think** : 7,000 thousands = 7,000,000]

check your understanding

How does the value of 5 change as it moves from the Hundreds place to the Thousands place ?

- Let your child understand that in a multi-digit whole number, a digit in one place represents ten times what it represents in the place to its right.

Exercise

1

on lessons 1&2

► Big Numbers! ► Changing Values

REMEMBER UNDERSTAND APPLY PROBLEM SOLVING

From the school book

1. Complete the table as in the example.

		MILLIARDS			MILLIONS			THOUSANDS			ONES		
	Number	O	H	T	O	H	T	O	H	T	O		
Ex.	5,604,453,987	5	6	0	4	4	5	3	9	8	7		
a.	8,714,326,518												
b.	753,009,300												
c.	7,354,621												
d.	8,000,300												
e.	923,508												

2. In the number 1,542,345,678 , What digit is in the

- | | |
|--------------------------|------------------------------|
| a. Tens place ? | b. Hundreds place ? |
| c. One Thousands place ? | d. Hundred Thousands place ? |
| e. Ten Millions place ? | f. One Billions place ? |

3. Write the value of the underlined digit according to its place in each number as in the example.

► Example: 47,209,531 → 40,000,000

- | | |
|---------------------------|-----------------------------|
| a. 58, <u>4</u> 86,098 → | b. 3, <u>7</u> 84,168,411 → |
| c. <u>6</u> 2,478,300 → | d. 462, <u>4</u> 17 → |
| e. 2 <u>4</u> ,041,683 → | f. 8,00 <u>0</u> ,418,617 → |
| g. 41, <u>6</u> 91,403 → | h. <u>3</u> 21,428,218 → |
| i. 669, <u>0</u> 84,422 → | j. 7,261, <u>9</u> 09 → |
| k. 30,30 <u>3</u> ,333 → | l. <u>6</u> 1,230,478 → |

[El-Monofia - Shebin El Koum 22]

4. Complete.

- a. The place value of the digit 2 in the number 2,500,000 is _____ [Souhag 23]
- b. The place value of the digit 3 in the number 1,365,854 is _____ [Giza - Abo El-Nomros 23]
- c. The place value of the digit 3 in the number 23,174,265 is _____ [Giza - 6th October 22]
- d. The place value of the digit 9 in the number 91,300,122 is _____ [Alex. - Agmy 24]
- e. The place value of the digit 0 in the number 5,321,041,758 is _____
- f. The value of the digit 5 in the number 346,251,813 is _____ [Cairo 23]
- g. The value of the digit 6 in the number 26,715,324 is _____ [Cairo - Math's Inspection 23]
- h. The value of the digit 6 in 61,230,478 is _____ [Kafr El-Sheikh 24]
- i. The value of 2 in the Tens place = _____
- j. The value of 7 in the Hundreds place = _____
- k. The million is the smallest number formed from _____ digits.
- l. The milliard is the smallest number formed from _____ digits.
- m. The smallest number formed from the digits 9,7,6,8,3,1 and 4 is _____
- n. The smallest number formed from the digits 4,2,6,0,7,5,1 and 3 is _____
- o. The greatest number formed from the digits 8,1,3,4,5,0,9,7 and 2 is _____
- p. The greatest number formed from 7 digits is _____

5. Complete.

- a. 75,421,392 = _____ million, _____ thousand, _____
- b. 2,500,422,300 = _____ milliard, _____ million, _____ thousand, _____
- c. _____ = 701 million, 7 thousand, 700
- d. _____ = Two million, five hundred thousand, four hundred twenty-two.
- e. _____ = 9 milliard, 9 million, 9 thousand, 9
- f. _____ = 27 million, 27 thousand.
- g. _____ = Sixteen milliard, sixteen.

6. Fill in the blanks as in the example.

- Ex. • 23,800 = 238 hundreds.
• 60 tens = 600

75,000

- 750 hundreds = 75 thousands.

a. 56,000 = _____ thousands.

b. 280,000 = _____ hundreds.

c. 32,000 = _____ thousands.

[El-Kalyoubia - Math's Inspection 23]

d. 300 thousands = _____ hundreds.

e. 55 thousands = _____ hundreds.

f. 72,000 tens = _____ thousands.

g. 87,900,000 hundreds = _____ millions.

h. 80 tens = _____ [El-Menia 23]

i. 30 tens = _____ [Cairo - El-Shrouk 23]

j. 40 tens = _____ [Cairo 23]

k. 60 thousands = _____

l. The value of 50 thousands = _____ [Alexandria - Montaza 23]

7. Amir says that in the number 222, all of the digits have the same value.

Do you agree or disagree? Use words and numbers to explain your thinking.

8. Fill in the blanks.

a. _____ is 10 times one hundred thousand.

b. _____ is 10 times two hundred.

c. _____ is 10 times seven thousand.

d. Hundred thousand is _____ times ten thousand.

Challenge

9. Use the digits 5, 7, 3, 1, 8, 2, 9 and 6 to make the greatest number you can, then use the same digits to make the smallest number you can.

- The greatest is _____
- The smallest is _____
- How did the value of 7 change from the greatest number and the smallest number? Why did it change? Use words and numbers to explain your thinking.





Choose the correct answer.

1. The digit in the Ten Thousands place in the number 1,351,278 is _____. [Luxor 24]
 • A. 3 B. 2 C. 5 D. 1

2. The digit in the Hundred Thousands place in the number 3,910,472 is _____. [El-Beheira 24][El-Monofia - Al-Shohdaa 24]
 • A. 1 B. 2 C. 4 D. 9

3. Million is the smallest number formed from _____ digits. [Qena 24][Ismailia 24][El-Monofia - Berket El-Sabaa 24]
 • A. 5 B. 6 C. 8 D. 7

4. Milliard is the smallest number formed from _____ digits. [El-Menia 24]
 • A. 7 B. 8 C. 9 D. 10

5. The place value of the digit 7 in the number 7,213,455,686 is _____. [Kafr El-Sheikh 24]
 • A. Millions B. Ten Millions C. Millions D. Ten Thousands

6. The value of the digit 6 in the Ten Millions place equals _____. [Kafr El-Shiekh 24]
 • A. 60,000 B. 6,000,000 C. 60,000,000 D. 600,000,000

7. The value of digit 7 in number 7,125,801 is _____. [El-Beheira - Maths' Inspection 23]
 • A. 7 B. 70 C. 7,000 D. 7,000,000

8. The value of digit 6 in number 2,476,217 is _____. [El-Menia Official Lang. School 23]
 • A. 60 B. 600 C. 6,000 D. 60,000

9. The value of the digit 0 in the number 301,572,941 is _____.
 • A. 0 B. 100 C. 1,000 D. 100,000

10. The value of the digit 5 in the number 2,456,300 is _____.
 • A. 5 millions. B. 50 millions. C. 50 thousands. D. 500 thousands.

11. The period of the underline digits in the number 25,613,729,114 is _____.
 • A. Millions. B. Millions. C. Thousands. D. Ones.

12. In which number does the 8 have a value of eight hundred ?
 • A. 538,419 B. 781,015 C. 271,825 D. 419,782

13. The number in which the digit 7 has the greatest value is _____.
 • A. 821,730,521 B. 152,007,000 C. 51,078,623 D. 7,810,521

14. In the number 34,042, the digit 4 in the Thousands place is equal to _____ times the digit 4 in the Tens place. [Cairo - Maths Inspection 22]
 • A. 10 B. 100 C. 1,000 D. 10,000

15. If a digit moves one space to the left on the place value chart, its value equals _____ times.
 • A. 1 B. 10 C. 100 D. 1,000

Lessons 3 & 4

- Many Forms to Write Numbers
- Composing and Decomposing

Learn 1 Many ways to write numbers

The distance between Jupiter
and the sun is about 778,340,821 km.



Place-Value Chart								
MILLIONS			THOUSANDS			ONES		
H	T	O	H	T	O	H	T	O
7	7	8	3	4	0	8	2	1

Standard Form : 778,340,821

"Commas are used to show periods"

Expanded Form : 700,000,000 + 70,000,000 + 8,000,000
+ 300,000 + 40,000 + 800 + 20 + 1

"Zeroes are not needed in expanded form because there is nothing in that place value; as 0 in Thousands place".

Word Form : Seven hundred seventy-eight million, three hundred forty thousand, eight hundred twenty-one.

"Commas are used to separate Millions, Thousands and Ones periods".

Short-Word Form : 778 million,340 thousand,821.

Notes

- We use standard form most often.
- Numbers written in expanded form show the value of each digit.

Notes for parents :

- Your child may be confused about how to represent a place value with a 0 digit in expanded form. For example : 30,456 = 30,000 + 400 + 50 + 6. The 0 is not represented in expanded form because in standard form the 0 represents that there is nothing in that place value.

Learn 2 Composing and decomposing numbers

- **Composing** numbers means [put together], and **decomposing** numbers means [broken apart].
- You can decompose the number **5,456,387** in different ways using place-value chart :

MILLIONS			THOUSANDS			ONES		
H	T	O	H	T	O	H	T	O
		5	4	5	6	3	8	7

► 1st way : Expanded Form :

$$5,456,387 = 5,000,000 + 400,000 + 50,000 + 6,000 + 300 + 80 + 7$$

► 2nd way :

$$5,456,387 = [5 \times 1,000,000] + [4 \times 100,000] + [5 \times 10,000] + [6 \times 1,000] + [3 \times 100] + [8 \times 10] + [7 \times 1]$$

Example 4

Complete the following.

a. Composed : 8,035,402,176

Decomposed : _____

b. Composed : _____

Decomposed : $[7 \times 1,000,000] + [9 \times 100,000] + [8 \times 1,000] + [2 \times 10] + [5 \times 1]$

Solution

a. Decomposed : $[8 \times 1,000,000,000] + [3 \times 10,000,000] + [5 \times 1,000,000] + [4 \times 100,000] + [2 \times 1,000] + [1 \times 100] + [7 \times 10] + [6 \times 1]$

b. Composed : 7,908,025

check your understanding

Complete the following.

a. Composed : 7,504,092,415

Decomposed : _____

b. Composed : _____

Decomposed : $[3 \times 1,000,000,000] + [2 \times 100,000,000] + [5 \times 10,000,000] + [4 \times 100,000] + [7 \times 10,000] + [8 \times 1,000] + [6 \times 10] + [9 \times 1]$

Notes for parents :

- Make sure that your child knows the difference between the terms compose and decompose.
- Make sure that your child knows how to represent a zero in a place when the number is decomposed.

Exercise

2

on lessons 3&4

- Many Forms to Write Numbers
- Composing and Decomposing

REMEMBER

UNDERSTAND

APPLY

PROBLEM SOLVING

From the school book

Many forms to write numbers

1. Write each number in standard form.

- a. Four hundred and nine. _____ [El-Monofia - Quesna 22]
- b. 34 million, 97 thousand. _____ [Giza - Kerdasa 22]
- c. Three million, two hundred fourteen thousand, nine hundred thirty-six.
_____ [El-Menia - Samalot 22]
- d. Five hundred twenty-seven million, nine hundred thousand, six hundred forty.

- e. Three milliard, four hundred two million, seventeen. _____
- f. $20,000 + 7,000 + 400 + 20 + 2$ _____
- g. $70,000,000 + 126,000 + 450$ _____ [El-Menia - Der Mawas 22]

2. Write the expanded form of each of the following.

- a. 1,756,300 _____
- b. 54,632,405 _____
- c. 701,462,051 _____
- d. 9,989,791,985 _____
- e. 35 million, 17 thousand, 230 _____
- f. Two milliard, four hundred twenty million, three hundred fifty-two thousand, one hundred three _____

3. Write each number in word form.

- a. 3,562,504 _____
- b. 54,213,450 _____
- c. 5,408,921,002 _____

d. $700,000 + 60,000 + 20 + 9$ _____

e. $5,000,000,000 + 7,000,000 + 900,000 + 3,000 + 20$ _____

4. Complete the following.

a. The standard form of the number "3 million, 21 thousands and 509" is _____ [Alex. - Agmy 24]

b. The standard form of the number "one million, twenty-four thousand" is _____ [Alex. - Agmy 24]

c. The standard form of the number "6 million, 221 thousand" is _____ [Alex. - Agmy 24]

d. 8 million, 555 thousand, 666 = _____ [Cairo 24]

e. 5 milliard, 5 thousand and 5 = _____ [El-Menia 24]

f. 4 milliard, 25 million, 67 thousand, 59 = _____ [Kafr El-Sheikh 24]

g. Three millions, three thousands, three in standard form is _____ [Giza 24]

h. Five million, six hundred fifty thousand and sixteen is _____ [Ismailia 24]

i. $5,000,000 + 20,000 + 3,000 + 600 + 40 =$ _____ [standard form]. [Alex. 24]

j. $700,000 + 40,000 + 2,000 + 300 + 70 + 2 =$ _____ [El-Menia - Malawi 24]

k. $4 + 500 + 3,000 + 800,000 =$ _____ [El-Monofia - Berket El Sabaa 24]

l. The number 6 million, 543 thousand, 210 in standard form is _____ [Giza - October Gardens 24]

m. The standard form of the number 1 million, 235 thousand and 789 is _____ [Souhag 23]

n. $7,625 = 5 + 7,000 + 20 +$ _____ [Aswan 23]

o. $3,000,000 + 8,000 + 400 + 30 + 3 =$ _____ [Alexandria - First Montaza 23]

Compsing and decomposing numbers

5. Decompose the following numbers using expanded form.

a. $170,392 =$ _____ + _____ + _____ + _____ + _____

b. $105,208 =$ _____ + _____ + _____ + _____

c. $601,207 =$ _____ [El-Menia 2022]

d. 2 million, 277 thousand, 191 = _____

e. 17 million, 230 thousand, 14 = _____

f. Three milliard, one hundred thirty-seven million, six hundred nineteen thousand, eighty-eight = _____

6. Fill in the missing numbers. Use the place-value chart to help you.

a. Composed : 6,124,030,420

Decomposed :

MILLIARDS	MILLIONS			THOUSANDS			ONES		
O	H	T	O	H	T	O	H	T	O
_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

b. Composed :

Decomposed :

MILLIARDS	MILLIONS			THOUSANDS			ONES		
O	H	T	O	H	T	O	H	T	O
5	4	0	0	1	5	9	0	2	4

c. Composed :

Decomposed : $[7 \times 1,000,000,000] + [5 \times 10,000,000] + [4 \times 10,000] + [3 \times 1,000]$
 $+ [5 \times 100] + [9 \times 1]$

MILLIARDS	MILLIONS			THOUSANDS			ONES		
O	H	T	O	H	T	O	H	T	O
_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

7. Complete the table.

	Composed	Decomposed
a.	4,040,400	_____
b.	_____	$[2 \times 1,000,000,000] + [5 \times 1,000,000] + [6 \times 100,000]$ $+ [7 \times 1000] + [9 \times 100] + [2 \times 10] + 7$
c.	_____	$9,000,000 + 50,000 + 3,000 + 700 + 60 + 9$
d.	9,210,031,458	_____

Challenge

8. Write 16 ten thousands + 5 thousands + 64 tens in standard form. _____

9. Find two 9-digit numbers with the difference between them is one million.

_____ and _____



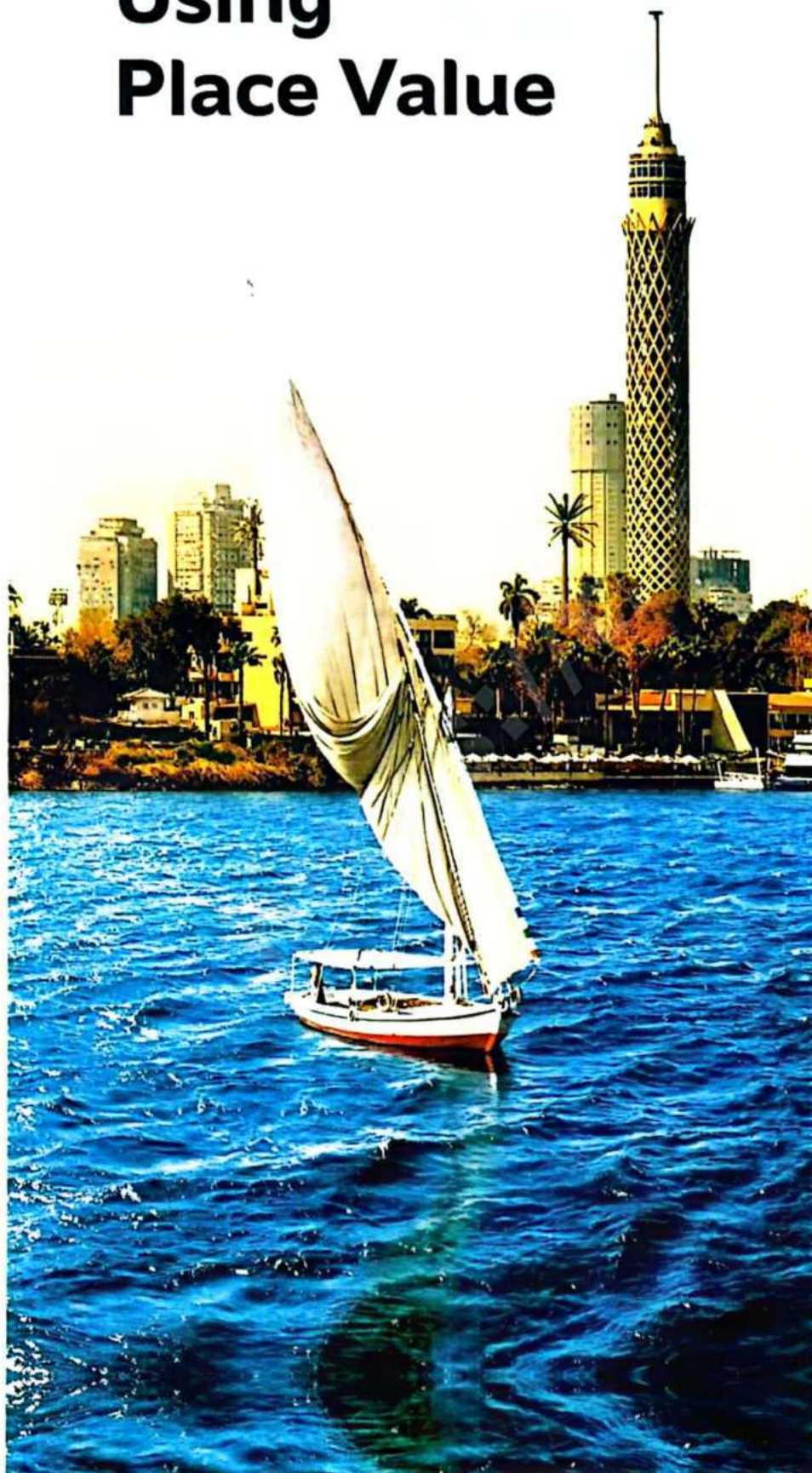
Multiple Choice Questions

Choose the correct answer.

1. The number building of the number :
9,231,043,204 is called _____ form.
[Aswan 23]
A. decomposed B. standard
C. expanded D. word
2. The number 10 million, 175 thousand, 314 is
written in the standard form _____.
[El-Monofia - Al-Shohdaa 24]
A. 10,157,314 B. 10,571,413
C. 10,175,314 D. 10,751,314
3. What is the standard form of eighteen
million, six hundred five thousand ?
[Alexandria - El-Montaza 22]
A. 18,605,000 B. 81,605,000
C. 1,860,500 D. 18,650,000
4. The standard form of 5 million, 36 thousand
and 206 is _____.
[Cairo 23]
A. 5,000,036,206 B. 5,036,206
C. 532,206 D. 5,360,206
5. The number 2 million, 300 thousand in
standard form is _____. [El-Kalyoubia 23]
A. 2,300,000 B. 2,000,300
C. 2,300 D. 2,003,000
6. Nine millions and six hundreds = _____.
[Souhag 24]
A. 600,900 B. 900,600
C. 960,000 D. 9,000,600
7. The number 1 milliard, 235 million, 127 in
standard form = _____. [Cairo 24]
A. 1,235,000,127 B. 1,235,127
C. 1,272,351 D. 1,235,127,000
8. $9,000,000 + 6,000 + 50 + 6 =$ _____.
[Port Said 24]
A. 9,656 B. 960,666
C. 6,569 D. 9,006,056
9. $5,000,000 + 40,000 + 5,000 + 600 + 30$
 $+ 4 =$ _____. [Cairo - Misr El Kadima 24]
A. 5,456,304 B. 5,045,634
C. 5,405,634 D. 504,534
10. The number 173 million, 904 thousand, 562
in standard form is _____. [El-Dakahlia 22]
A. 173,000,904,562 B. 173,940,562
C. 173,904,562 D. 173,562,904
11. The number 309,602 [in expanded form]
is _____. [Port Said 24]
A. $2 + 60 + 900 + 3,000$
B. $2 + 600 + 9,000 + 30,000$
C. $2 + 600 + 9,000 + 300,000$
D. $2 + 60 + 9,000 + 300,000$
12. $[3 \times 1,000] + [3 \times 10] =$ _____. [El-Menia 24]
A. 300 B. 3,030
C. 3,300 D. 30,030
13. The composed number of $[7 \times 100,000] + [2 \times 1,000] + [3 \times 100] + [5 \times 1]$ is _____.
[Alex. - Agmy 24]
A. 720,351 B. 702,305 C. 72,305 D. 702,350

CONCEPT 2

Using Place Value



► Lessons 5&6

Comparing Big Numbers
Comparing Numbers in Multiple Forms

Learning Objectives:

- Students will use place value to compare large numerals.
- Students will use symbols to express numerical comparisons.
- Students will compare numbers in multiple forms.
- Students will describe strategies for comparing numbers in multiple forms.

► Lesson 7

Descending and Ascending Numbers

Learning Objectives:

- Students will order numbers in multiple forms.
- Students will describe strategies for ordering numbers in multiple forms.

► Lesson 8

Rounding Rules

Learning Objectives:

- Students will apply multiple strategies to round numbers.
- Students will discuss whether rounding or front-end estimation provide a more accurate estimate.

Fast Fact

The Nile River is the longest river in the world. It has a length about **[6,659 kilometers]**. Compare its length with the Amazon. The Amazon is about **[6,400 kilometers]** long.

Lessons 5 & 6

- ▶ Comparing Big Numbers
- ▶ Comparing Numbers in Multiple Forms

Learn How to compare numbers ?

First Comparing numbers have different numbers of digits

When comparing numbers, the number which has more digits is the greater.

For Example : $5,302,200 > 899,529$

because 5,302,200 has more digits than 899,529

Second Comparing numbers have the same number of digits

- You can compare two numbers with the same number of digits **by starting at the left and moving right** until you come to a pair of digits that **do not have the same value**.

For Example :

To compare 12,673 and 12,763. Start at the left. Check each place until the digits are different.

Step 1

Compare the Ten Thousands.

$12,673$
↓ same digit of Ten Thousands
 $12,763$

Step 2

Compare the Thousands.

$12,673$
↓ same digit of Thousands
 $12,763$

Step 3

Compare the Hundreds.

$12,673$
↓ $7 > 6$
 $12,763$

Then $12,763 > 12,673$

More Examples :

• $754,042 < 755,950$

• $755,972 < 1,752,421$

• $42,437 > 42,347$

• $6,406,367 = 6,406,367$

Notes for parents :

- Ask your child to consider how many digits are in a number when he/she compares.

Example 1

Write [$>$, $<$ or $=$] to compare.

a. 37,048 37,184

c. 4,010,065 4,000,056

b. 217,906 271,906

d. 810,340 810,340

Solution

a. 37,048 $<$ 37,184

c. 4,010,065 $>$ 4,000,056

b. 217,906 $<$ 271,906

d. 810,340 $=$ 810,340

Example 2

Complete.

a. The smallest number formed from 3, 7, 1, 9, 2, 6, 5, is _____

b. The smallest number formed from 8, 5, 0, 1, 3, 9 is _____

c. The greatest number formed from 4, 2, 1, 3, 7, 6, 5 is _____

d. The greatest number formed from 5, 0, 6, 2, 1, 7, 4 is _____

Solution

- To create the greatest number, arrange the digits from greatest to least.
- To create the smallest number, arrange the digits from least to greatest.

a. 1,235,679

b. 103,589

c. 7,654,321

d. 7,654,210

Note

If the digits contains "Zero" you can't put it on the left of the number.

Check your understanding

Write [$>$, $<$ or $=$] to compare.

a. 2,346 2,338

c. 723,215 723,215

e. 503,278,105 503,279,100

b. 478,765 479,112

d. 752,321,271 72,321,271

f. 7,492,102,235 7,491,102,235

Notes for parents :

- If your child has difficulty making comparisons, let him/her first circle the place where the digits are different.

Learn How to compare numbers in multiple forms ?

- You can compare numbers in any forms : standard, expanded and word form.
- You may convert to standard form to compare, or use place value in expanded form or in word form to compare.

Example 3

Write [$>$, $<$ or $=$] to compare.

- | | | | |
|----|--|-----------------------|---|
| a. | $70,000 + 4,000 + 50 + 7$ | ☐ | $70,000 + 4,000 + 500 + 70$ |
| b. | Two milliard, seven hundred thirty-eight thousand, ten. | ☐ | Two milliard, seven hundred thirty-five thousand, eleven. |
| c. | $3,000,000 + 7,000 + 800 + 9$ | ☐ | Three million, seven thousand, eight hundred nine. |
| d. | $[7 \times 1,000,000] + [5 \times 100,000] + [3 \times 1,000] + [4 \times 100] + [9 \times 1]$ | ☐ | $7,000,000 + 500,000 + 3,000 + 400 + 90$ |

Solution

- a. $<$ b. $>$ c. $=$ d. $<$

Check your understanding

Write [$>$, $<$ or $=$] to compare.

- | | | | |
|----|--|-----------------------|---|
| a. | $500,000 + 70,000 + 90 + 8$ | ☐ | $1,000,000 + 5,000 + 1$ |
| b. | Three milliard, two hundred fifty-two thousand, three hundred four. | ☐ | Three milliard, two hundred fifty-two thousand, thirty-four. |
| c. | $[8 \times 1,000,000] + [6 \times 1,000] + [5 \times 100] + [7 \times 10]$ | ☐ | $[8 \times 1,000,000] + [2 \times 10,000] + [6 \times 1,000] + [5 \times 100] + [9 \times 1]$ |
| d. | $2,000,000,000 + 400,000,000 + 2,000 + 30 + 2$ | ☐ | $2,000,000,000 + 50,000,000 + 8,000,000 + 9,000 + 50 + 9$ |

- Your child may struggle with comparing numbers in word form or expanded form. He/she may convert to standard form to compare.

Exercise 3

on lessons 5&6

► Comparing Big Numbers

► Comparing Numbers in Multiple Forms

● REMEMBER ● UNDERSTAND ● APPLY ● PROBLEM SOLVING

From the school book

1. Compare. Write ($>$, $<$ or $=$).

a. 707

☐

770

b. 1,207

☐

1,207

c. 10,525

☐

10,255

d. 190,098

☐

19,098

e. 123,568

☐

123,978

f. 6,235,678

☐

6,235,508

g. 2,450,890,007

☐

2,500,000,000

h. 7,798,562,415

☐

7,798,567,999

i. 89,418,147

☐

89,418,247

j. 571,600,254

☐

571,600,329

k. 1,000,000,000

☐

900,000,000

l. 100,000,000

☐

99,999,999

2. Compare. Write ($>$, $<$ or $=$).

a.

40,000

☐

400 thousands

b.

7 ten thousands

☐

7,000

c.

7,000 millions

☐

7 milliards

d.

5 milliards, 367 thousands

☐

5,367,000,000

e.

Ninety-seven million, three hundred one

☐

$90,000,000 + 7,000,000 + 3,000 + 1$

f.

5,193,492,500

☐

Five milliard, three hundred million, seven hundred fifteen thousand, forty-three.

g.

$(7 \times 100,000,000) + (4 \times 10,000,000) + (9 \times 10,000) + (8 \times 100) + (1 \times 10)$

☐

$70,000 + 9,000 + 600 + 40 + 3$

h.

8,040,761,903

☐

$8,000,000,000 + 400,000,000 + 700,000 + 60,000 + 1,000 + 900 + 3$

- i. Seventeen million, four hundred twenty-five thousand, six hundred five. 17,420,605
- j. Four hundred twenty-three thousand, twelve $400,000 + 30,000 + 2,000 + 20 + 1$

3. Complete.

- a. The smallest number formed from 8, 2, 9, 0, 5, 1, 7 = _____ [Souhag 24]
- b. The smallest number formed from the digits 2, 0, 5, 3 is _____ [El-Monofia - Menof 24]
- c. The smallest number formed from the digits 6, 0, 9, 8, 4, 2 and 3 is _____ [Port Said 24]
- d. The greatest number consists of the digits 9, 5, 0, 2, 8 is _____ [Cairo - Helwan 24]
- e. The greatest number formed from the digits 3, 7, 1, 0, 4, 2 is _____ [Port Said 24]

4. Find each missing digit.

- | | | |
|-------------------------------|--------------------------------|--------------------------------|
| a. $6,106 > 6 \square 19$ | b. $2,117 = \square,117$ | c. $4,382 < 4,3 \square 2$ |
| d. $91,472 > 9 \square,472$ | e. $114,899 < 114, \square 99$ | f. $703,9 \square 1 = 703,981$ |
| g. $11,234 > 1 \square,785$ | h. $67,813 > 67,8 \square 3$ | i. $82, \square 88 = 82,588$ |
| j. $179,00 \square < 179,001$ | k. $856, \square 34 < 856,134$ | l. $683,129 < 6 \square 3,129$ |

5. Write a number.

- a. Create a number that is less in the Hundred Thousands place than $[<] 893,820$
- b. Create a number that is greater in the Millions place than $[>] 178,462,490$ [Cairo - Heliopolis 23]
- c. Create a number that is less in the Ten Millions place than $[<] 32,427,400$
- d. Write a number in expanded form that is equal to $[=] 2,445,232,197$

 Challenge

6. Describe the error in the following number sentence, and then explain how you would correct it. $24,152,614 < 24,125,614$
7. Which is greater, the number that is 1,000 less than 13,495 or the number that is 10,000 less than 23,495?

Multiple Choice Questions

Choose the correct answer.

- | | |
|---|--|
| <p>1. 62,234 ————— 62,324 [Alex. - Al-Agamy 23]</p> <p>A. = B. > C. <</p> | <p>2. 1,248,056 ————— 998,578 [El-Monofia - Sadat City 23]</p> <p>A. > B. < C. =</p> |
| <p>3. 6,235,678 ————— 6,235,508 [Beni Suef 24]</p> <p>A. > B. < C. = D. ≤</p> | <p>4. 2,450,890 ————— 2,500,000 [Alex. 24]</p> <p>A. < B. > C. = D. ≥</p> |
| <p>5. 3,230,765 ————— 1,987,374 [Souhag 23]</p> <p>A. < B. = C. > D. ≤</p> | <p>6. 44,000,271 ————— 4,000,271 [Cairo 24]</p> <p>A. > B. < C. = D. non of them</p> |
| <p>7. 45,254,369 ————— 45,678,124 [Alex. 23]</p> <p>A. > B. = C. < D. ≥</p> | <p>8. 55,000,888 ————— 51,999,777 [Souhag 24]</p> <p>A. < B. > C. = D. ≤</p> |
| <p>9. 3,000 thousand ————— 9 hundred thousand [Alex. - Agmy 24]</p> <p>A. > B. < C. =</p> | <p>10. 501 thousand ————— 3,000 thousand [Alex. - Agmy 24]</p> <p>A. > B. < C. =</p> |
| <p>11. 9 million ————— 8,978,269 [El-Monofia - Menof 24]</p> <p>A. < B. = C. > D. Otherwise</p> | <p>12. 6 million, 400 thousand ————— 6,040,000 [El-Monofia 24]</p> <p>A. < B. = C. > D. Otherwise</p> |
| <p>13. Which of the following statements is TRUE ? [Cairo - Heliopolis 23]</p> <p>A. $4646 < 4664$ B. $4646 > 4664$</p> <p>C. $4664 > 4664$ D. $4646 = 4664$</p> | <p>14. Which number sentence is NOT TRUE ?</p> <p>A. $2,304 < 2,340$</p> <p>B. $27,920 > 27,790$</p> <p>C. $1,005,301 > 1,050,901$</p> <p>D. $80,044 < 80,404$</p> |

Lesson 7

Descending and Ascending Numbers

Learn Ordering numbers

The table shows the population of four governorates in Egypt in 2021. You can order the governorates by their population from greatest to least as follows :



1. The number 10,058,942 has the most number of digits, so it is the greatest number.

2. The number 450,528 has the least number of digits, so it is the least number.

3. Compare 5,452,718 and 5,510,876 which have the same number of digits. Check each place until the digits are different.

Governorate	Population
Cairo	10,058,942
North Sinai	450,528
Alexandria	5,452,718
Souhag	5,510,876

Step 1

Compare the Millions.

5,452,718
↓
5,510,876

same digit
of Millions

Step 2

Compare the Hundred Thousands.

5,452,718
↓
5,510,876

} 5 > 4

Then : 5,510,876 > 5,452,718

From above :

10,058,942 > 5,510,876 > 5,452,718 > 450,528

- In order of their population, the governorates are Cairo, Souhag, Alexandria and North Sinai.

Notes for parents :

- Remind your child to start comparison at the greatest place value.

Example 1

Write these numbers in an ascending order.

2,896,016

1,188,580

2,517,550

Solution

Step 1 → Write the numbers, lining up places. Determine the smallest number.

2,896,016

1,188,580 ← smallest

2,517,550

Step 2 → Write the remaining numbers, lining up places. Compare.

2,896,016

2,517,550 ← smaller

Step 3 → Write the numbers from least to greatest.

1,188,580

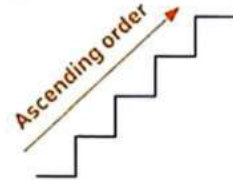
2,517,550

2,896,016



Remember

Ascending order is ordering numbers from least to greatest.



Example 2

Write each of the following numbers in standard form, then arrange them in a descending order.

- $[7 \times 1,000,000,000] + [4 \times 10,000,000] + [5 \times 1,000] + [3 \times 100]$
- Seven milliard, four hundred million, one thousand, two
- $7,000,000,000 + 500,000,000 + 600,000 + 300$
- 745,300

Solution

Standard form	Descendingly
7,040,005,300	7,500,600,300
7,400,001,002	7,400,001,002
7,500,600,300	7,040,005,300
745,300	745,300



Remember

Descending order is ordering numbers from greatest to least.



Check your understanding

Arrange the following in a descending order, using the standard form.

- Three milliard, forty million, seventy-one thousand, ten.
- $[3 \times 1,000,000,000] + [5 \times 1,000,000] + [7 \times 1,000] + [1 \times 100] + [1 \times 10]$.
- $3,000,000,000 + 30,000,000 + 10$
- 3,300,710,400

Notes for parents :

- If your child has trouble ordering numbers, ask him/her to align the numbers vertically and compare digits from left to right.
- Remind your child about the meaning of the two terms ascending order and descending order.

Exercise

4

on lesson 7

Descending and Ascending Numbers

REMEMBER UNDERSTAND APPLY PROBLEM SOLVING

From the school book

1. Write the numbers in an ascending order.

- a. 8,092,561 , 9,208,111 , 7,534,786 , 8,650,336

[El-Beheira 23]

The order is : _____ , _____ , _____ , _____

- b. 1,282,756 , 3,012,427 , 988,423 , 3,105,338

[El-Monofia - Sadat city 23]

The order is : _____ , _____ , _____ , _____

- c. 430,000,459 , 43,000,549 , 403,000,456 , 430,549,000

The order is : _____ , _____ , _____ , _____

2. Write the numbers in a descending order.

- a. 450,321 , 504,321 , 321,405 , 342,150 , 540,312

[Giza 23]

The order is : _____ , _____ , _____ , _____ , _____

- b. 6,562,942,735 , 6,942,735 , 6,562,942,375 , 6,942,537

The order is : _____ , _____ , _____ , _____

- c. 4,237,651 , 4,273,653 , 495,627 , 4,237,690

The order is : _____ , _____ , _____ , _____

3. List the following data in a descending order. You may use word or standard form.

- Three milliard, ten million, one thousand, thirty-four.
- Three milliard, one million, three hundred twenty-three thousand, three hundred ninety-one.
- Three milliard, nine hundred ninety thousand, nine hundred ninety-two
- Three milliard, one hundred ten million, ninety-nine thousand, four hundred ninety-three.

4. List the following in an ascending order. Use standard form.

- a. • 654,301 • Six hundred fifty-four thousand, three hundred ten.
 • 604,320 • 654,311
 • Five hundred ninety-nine thousand, three hundred ten.

The order is :

- b. •
- $3,000,000 + 400,000 + 5,000 + 3$

- Three million, four hundred fifty thousand, three.
 • 3,453,000 • 3,450,030

The order is :

5. List the numbers in a descending order. Use standard form.

- a. • Two milliard, four million, seven hundred thousand.
 • 2,400,700,000 • 2,040,007,000 • Three milliard.

The order is :

- b. •
- $5,000,000,000 + 40,000,000 + 5,000,000 + 7,000 + 90$

- $[6 \times 1,000,000,000] + [3 \times 10,000,000] + [5 \times 1,000,000] + [6 \times 10,000] + [9 \times 100]$
 • Five milliard, forty-one million, seven thousand, ninety
 • $6,000,000,000 + 40,000,000 + 5,000,000 + 10,000 + 7,000 + 90$
 • 6,025,060,990

The order is :

6. List the numbers in an ascending order. Use the form in which they are given.

a.  • Four milliard, six hundred thousand, four.

• 461,014

• Four milliard, six hundred thousand forty.

• $[4 \times 1,000,000,000] + [4 \times 100,000] + [6 \times 10]$ • 6,400,042

The order is : _____

b. • Nine million, seven hundred thirty-one thousand, seventy.

• 90,731,007

• 900,080,500

• $9,000,000 + 700,000 + 40,000 + 50$

• Seven hundred million, eighty-four.

The order is : _____

7. List the numbers in a descending order. Use the form in which they are given.

a. • 900 thousand.

• 9 million.

• 5 million and 7 hundred thousand.

• 550,223

[Giza - El-Haram 22]

The order is : _____

b. • Four milliard, six hundred thousand, four.

• 461,014

• Four milliard, six hundred thousand, forty.

• $[4 \times 1,000,000,000] + [4 \times 100,000] + [6 \times 10]$

• 6,400,042

The order is : _____

Challenge

8. The following numbers are arranged in a descending order :

3,751,924,096 $\longrightarrow$ 3,751,924,069 $\longrightarrow$ 3,751,624,096 $\longrightarrow$ 3,751,624,069

If you replace each 6 by 9 and each 9 by 6, what do you notice ?

Multiple Choice Questions

Choose the correct answer.

1. Which of the following shows the numbers in a descending order ?

A. 580 , 735 , 757 , 573 B. 735 , 508 , 573 , 757
C. 735 , 757 , 573 , 580 D. 757 , 735 , 580 , 573 [Giza - Awseem 22]

2. Which of the following is a correct ascending order ?

[Cairo - Heliopolis 23]

A. 757 , 573 , 508 , 735 B. 573 , 757 , 735 , 580
C. 573 , 580 , 735 , 757 D. 580 , 573 , 757 , 735

3. Which choice shows the numbers in a descending order ?

A. 1. 3,456,871	B. 1. 7,456,232	C. 1. 5,786,321	D. 1. 1,263,572
2. 3,578,462	2. 6,785,000	2. 5,795,786	2. 12,213,573
3. 987,541	3. 6,670,785	3. 5,895,432	3. 4,262,563
4. 5,743,261	4. 5,700,726	4. 6,721,000	4. 1,000,000,000
5. 8,784,561	5. 5,700,624	5. 7,000,000	5. 7,865,321,000

4. Given the following numbers :

a $[6 \times 100,000] + [4 \times 10,000] + [5 \times 1,000] + [3 \times 100] + [1 \times 1]$
b Six hundred fifty-three thousand, three hundred, ten.
c 604,302
d Five hundred eighty-eight thousand, three hundred, ten.

Which choice shows these numbers in an ascending order ?

A. a , c , b , d B. d , c , a , b C. d , b , a , c D. d , a , c , b

5. Which of the following digits makes the sentence true ? $785 > 7 \quad 5 > 755$

A. 2 B. 4 C. 6 D. 8

6. The table below shows the average distances from the planets to the Sun.

Planet	Jupiter	Mars	Venus	Earth
Distance from the Sun in km	778,340,821	227,943,000	108,209,000	149,598,000

Which planet from above is nearest to the Sun ?

A. Jupiter B. Mars C. Venus D. Earth

Lesson

8

Rounding Rules

Learn Different ways to round a number

A roller coaster that is 2,181 meters long. About how long is the roller coaster?

Since you do not need an exact number, you can **estimate** by **rounding** the number.



Different Ways to round 2,181 to the nearest thousand.

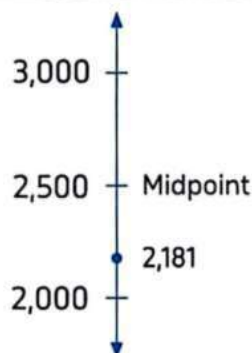
Way 1 You can use [midpoint strategy].

2,181 is closer to 2,000 than to 3,000

So, round 2,181 to 2,000

Written as $2,181 \approx 2,000$

and read as 2,181 approximately equals 2,000



Way 2 You can use [place value strategy]

Step 1

Find the place you want to round to. Circle the digit in that place.

2,181
↑
Thousands place

Step 2

Look at the digit to its right. Underline that digit.

2,181
↑
digit to the right

Step 3

- If the underlined digit is 5 or greater, round up.
- If the underlined digit is less than 5, round down.
- Change each digit to the right of the rounding place to 0
- 1 is less than 5, so

2,181 $\xrightarrow{\text{rounds to}}$ 2,000

Then, the roller coaster is about 2,000 meters long.

Notes for parents :

- Remind your child to round up if the digit to the right of the place value he/she wants to round to is equal to or greater than 5.

Example 1

Use the midpoint strategy to round each of the following.

a. 74,231 [to the nearest 1,000]

b. 9,360 [to the nearest 100]

Solution 

a. 74,231 is between 74,000
and 75,000

then

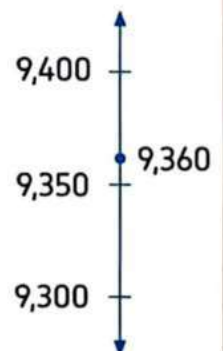
$$74,231 \approx 74,000$$



b. 9,360 is between 9,300
and 9,400

then

$$9,360 \approx 9,400$$



Example 2

Use the place value strategy to round each of the following.

a. 2,618 [to the nearest 10]

b. 174,568 [to the nearest 10,000]

c. 3,697,852,721 [to the nearest Ten Million]

d. 7,556,462 [to the nearest Million]

e. 999,999 [to the nearest Ten Thousand]

f. 13,999,999 [to the nearest Hundred]

Solution 

a. $2,6\text{1}8 \approx 2,620$ (8 > 5)

b. $174,5\text{6}8 \approx 170,000$ (4 < 5)

c. $3,6\text{9}7,852,721 \approx 3,700,000,000$ (7 > 5)

d. $7,5\text{5}6,462 \approx 8,000,000$ (5 = 5)

e. $9\text{9}9,999 \approx 1,000,000$ (9 > 5)

f. $13,9\text{9}9,999 \approx 14,000,000$ (9 > 5)

Rounding Rule :

Circle the digit, look next door.
5 or higher ? Add one more.
4 or less ? Let it rest.

Remember

The digits to the right become zeroes.

**ERROR
ALERT**

When you round a 9 to the next
greater digit, remember to
regroup to the next place value.

 **check** your understanding

Round the following.

a. 85,721 $\xrightarrow[\text{1,000}]{\text{to the nearest}}$

b. 3,562,291 $\xrightarrow[\text{Ten Thousand}]{\text{to the nearest}}$

c. 3,895 $\xrightarrow[\text{Hundred}]{\text{to the nearest}}$

d. 9,999,998 $\xrightarrow[\text{Million}]{\text{to the nearest}}$

Notes for parents :

- If necessary, allow your child to write the standard form of the number before rounding.

Exercise

5

on lesson 8

Rounding Rules

● REMEMBER ● UNDERSTAND ● APPLY ● PROBLEM SOLVING

From the school book

1. Round the numbers to the nearest Ten.

- a. $423 \approx$ _____ b. $549 \approx$ _____ c. $495 \approx$ _____ d. $1,287 \approx$ _____

2. Round the numbers to the nearest Hundred.

- a. $874 \approx$ _____ b. $416 \approx$ _____ c. $4,398 \approx$ _____ d. $1,952 \approx$ _____

3. Round the numbers to the nearest Thousand.

- a. $8,090 \approx$ _____ b. $234,432 \approx$ _____
c. $9,900 \approx$ _____ d. $7,578 \approx$ _____

4. Round the numbers to the nearest Ten Thousand.

- a. $37,205 \approx$ _____ b. $58,936 \approx$ _____
c. $290,290 \approx$ _____ d. $7,435,026,353 \approx$ _____

5. Round the numbers to the nearest Hundred Thousand.

- a. $483,267 \approx$ _____ b. $678,090 \approx$ _____
c. $449,300 \approx$ _____ d. $12,786,500 \approx$ _____

6. Round the numbers to the nearest Million.

- a. $5,367,544 \approx$ _____ b. $20,843,267 \approx$ _____
c. $135,984,600 \approx$ _____ d. $2,453,000,601 \approx$ _____

7. Round the numbers to the nearest Billion.

- a. $5,266,747,023 \approx$ _____ b. $10,944,352,543 \approx$ _____

8. Complete.

- a. $3,446 \approx$ _____ [to the nearest Thousand] [Cairo 24]
b. $56,621 \approx$ _____ [to the nearest Thousand] [Alex. - El-Montaza 24]
c. The number 543,186 to the nearest Thousand is _____ [Cairo - Heliopolis 22]
d. The number 163,518,943 to the nearest Million is _____ [Matrouh 22]
e. $34,279 \approx$ _____ [to the nearest Ten Thousand] [El-Monofia - Sadat City 23]
f. $4,369 \approx$ _____ [to the nearest 100] [Cairo - El-Nozha 23]
g. $16,401 \approx$ _____ [to the nearest Thousand] [Alex. - El-Montaza 23]

9. Round each of the following by using the midpoint strategy, record the midpoint of the number line and the place of each number, then round to the nearest Thousand.

a. $6,700 \approx$



b. $9,340 \approx$



c. $16,401 \approx$



d. $19,654 \approx$



10. Draw the number line, label the midpoint, and then round each of the following numbers.

a. $250,000$ [to the nearest Hundred Thousand]

b. $700,500$ [to the nearest Hundred Thousand]

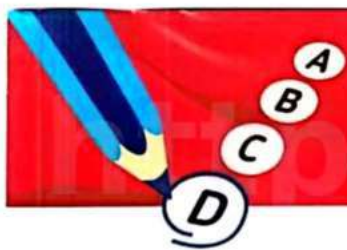
c. $362,261$ [to the nearest Ten Thousand]

d. $36,951$ [to the nearest Hundred]

11. Write 5 numbers if rounded to the nearest Thousand the result is 312,000.

Challenge

12. What is the greatest whole number that rounds to 300,000? What is the least?



Multiple Choice Questions

Choose the correct answer.

- | | |
|---|--|
| <p>1. $6,438 \approx 6,400$ [to the nearest]
[El-Menia - Dier Mawas 24]</p> <p>A. Unit B. Ten
C. Hundred D. Thousand</p> | <p>2. $25,638 \approx 26,000$ [rounding to the nearest]
[Giza 24]</p> <p>A. Ten thousand B. Thousand
C. Hundred D. Ten</p> |
| <p>3. $3,965,832 \approx 4,000,000$ to the nearest
[Cairo 24]</p> <p>A. Million B. Milliard
C. Hundred Thousand D. Ten Thousand</p> | <p>4. $2,357 \approx$ [rounding to the nearest Ten]
[Giza 23]</p> <p>A. 2,360 B. 2,358
C. 2,350 D. 2,400</p> |
| <p>5. When approximating the number 3,629 to the nearest Hundred, the result is
[Souhag 23]</p> <p>A. 3,600 B. 3,700
C. 3,000 D. 3,620</p> | <p>6. A bee hive contains 102,635 bees, the number of bees to the nearest Ten Thousand is
[El-Monofia - Berket El-Sabaa 23]</p> <p>A. 100,000 B. 10,000
C. 102,010 D. 12,090</p> |
| <p>7. $614,231 \approx$ [rounding to nearest Hundred Thousand]
[Alex. Agmy 24]</p> <p>A. 600,000 B. 700,000
C. 614,000 D. 4,000</p> | <p>8. Rounding the number 34,089 to the nearest Ten Thousand is
[Beni Suef 24][El-Sharkia 22]</p> <p>A. 34,000 B. 34,090
C. 30,000 D. 35,000</p> |
| <p>9. Rounding the number 3,854,125 to the nearest Million is
[Giza - October Gardens 24]</p> <p>A. 2,000,000 B. 3,000,000
C. 4,000,000 D. 5,000,000</p> | <p>10. Which answer represents rounding 32,582,346 to the nearest Million?
[Giza - Awseem 23][Suez 22]</p> <p>A. 30,000,000 B. 32,600,000
C. 32,000,000 D. 33,000,000</p> |
| <p>11. Which answer could be rounded to 120,000 when rounded to the nearest Ten Thousand?
[Suez 22]</p> <p>A. 125,678 B. 116,034
C. 112,625 D. 20,789</p> | <p>12. Round 6,749,001,551 to the nearest Milliard.
[Beni Suef 22]</p> <p>A. 6,000,000,000 B. 7,000,000,000
C. 6,700,000,000 D. 8,000,000,000</p> |

Unit One Assessment



1. Choose the correct answer.

1. The digit in ten thousands place in the number 6,387,512 is _____ [El-Menia 23]
 A. 3 B. 4 C. 7 D. 8
2. Milliard is the smallest _____ - digit number. [Cairo 23]
 A. 5 B. 10 C. 9 D. 8
3. The place value of the digit 6 in 56,724,033 is [El-Beheira - Math Inspection 23]
 A. Thousands. B. Hundred Thousand.
 C. Millions. D. Ten Million.
4. The value of the digit 3 in 53,496,752 is [Aswan 23]
 A. 30 B. 30,000 C. 3,000,000 D. 300,000
5. Rounding the number 34,089 to the nearest Ten Thousand is [Cairo - Heliopolis 23]
 A. 34,000 B. 34,090 C. 30,000 D. 35,000
6. The standard form of : three hundred thirty-two million, forty-five thousand, two hundred five is [Cairo - Helwan 24]
 A. 205,045,332 B. 332,045,205 C. 231,430,24 D. 231,043,042
7. 3,752,000 three milliard, twenty.
 A. > B. < C. =

2. Complete the following.

1. One million is the smallest number formed from _____ digits. [Aswan 23]
2. The smallest number formed from the digits 9, 4, 2, 6, 0, 5, 1 is [Giza 24]
3. The value of the digit 6 in 61,230,478 is _____ [Cairo - Misr El-Kadima 24]
4. The place value of 2 in the number 6,268,503 is [Kafr El-Sheikh 24]
5. $80,000,000 + 124,000 + 650 =$ _____
6. $735,462 \approx$ _____ [Rounded to the nearest Ten Thousand]
7. 3,504,800,501 in expanded form is _____
8. $5,856,469 \approx 5,900,000$ [Rounded to the nearest _____]

3. Choose the correct answer.

- Rounding 32,582,346 to the nearest Million equals _____ [Cairo - Misr El-Kadima - 24]
 A. 30,000,000 B. 32,600,000 C. 32,000,000 D. 33,000,000
- The largest 5-digit number is _____
 A. 10,000 B. 100,000 C. 99,999 D. 98,765
- 100,000 is _____ times the number 10,000
 A. 10 B. 100 C. 1,000 D. 10,000
- What is the standard form for three milliard, seven hundred thirty-five thousand, fifty ?
 A. 3,735,000,050 B. 3,735,500 C. 3,000,735,050 D. 3,735,050
- Rounding the number 765,017 to the nearest Hundred Thousand is _____ [Alex.-Al-Agamy 23]
 A. 770,000 B. 800,000 C. 700,000 D. 760,000
- $[6 \times 1,000,000] + [5 \times 100,000] + [3 \times 1,000] + [2 \times 100] =$ _____ [Port Said - 24]
 A. 2,305,600 B. 3,605,200 C. 6,305,200 D. 6,503,200
- The place value of the digit 3 in the number 5,316,725,891 is _____ [Ismailia 24]
 A. Millions B. Hundred Millions
 C. Ten Millions D. Hundred Thousands

4. Answer the following.

- A plane's altitude increased by 2,721 meters.
Round this number to the nearest Hundred.
- Use the digits 7,4,2,0,3,5,6,8 to make the greatest number you can.
Then use the same digits to make the smallest number you can and round each number to the nearest Million.
- Arrange the following numbers in: 6,362,012 / 2,265,698 / 13,561,954 / 5,364,569

[Giza - October Gardens 24]

- A descending order: _____, _____, _____, _____
- An ascending order: _____, _____, _____, _____

4. Compose and decompose the following number.

MILLIARDS	MILLIONS			THOUSANDS			ONES		
0	H	T	O	H	T	O	H	T	O
2	8	0	5	4	0	0	6	9	3

Composed : _____

Decomposed : _____